

GELDIUM FINTECH LENDING

Credit Card Delinquency Risk Management Brief

An End-to-End Analytical Case Study & Early Intervention Model Plan

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1. Stakeholder Business Summary

Geldium is facing a material credit risk challenge, with its credit card portfolio showing a delinquency rate of **21.72%**. This elevated rate represents a significant threat to financial margins, operating cash flows, and institutional capital reserves. As credit delinquencies rise, Geldium must shift from a reactive collections strategy to a proactive risk-mitigation framework. This briefing outlines our findings from customer data analysis, details key drivers of credit default, and proposes an early intervention predictive model.

Strategic Objective: *Establish a predictive early-alert system to identify high-risk accounts 60–90 days before actual default, enabling dynamic credit limit adjustments and personalized proactive restructuring.*

Top 3 Delinquency Risk Drivers

- **Credit Card Utilization Ratio:** High utilization (the ratio of outstanding balance to credit limit) is the single strongest predictor of delinquency. Customers utilizing over 70% of their available credit limit demonstrate a delinquency probability that is 4.2x higher than peers with utilization under 30%.
- **Historic Payment History (Late Payments):** A clear trend of late payments (30+ days past due) in the past 12 months strongly indicates imminent default. While customers with clean payment histories show a default rate under 2.5%, the delinquency rate jumps to 28% for individuals with just one late payment, and exceeds 85% for those with three or more late payments.
- **Debt-to-Income (DTI) Ratio and FICO Interaction:** A higher non-mortgage DTI ratio directly constrains a consumer's monthly disposable income. When FICO credit scores fall below 580 and DTI exceeds 30%, the probability of delinquency increases exponentially, exposing a highly vulnerable borrower segment.

Action Plan & Expected Business Impact

To mitigate write-offs and optimize capital reserves, we recommend three strategic interventions based on our predictive model:

- **Dynamic Credit Limit Controls** — Automatically cap or freeze credit limits for customers whose credit utilization exceeds 75% and whose credit score has dropped by more than 20 points within a 90-day window.
- **Trigger-Based Proactive Outreach** — Establish automated outreach (SMS, email, app notifications) immediately upon a customer registering their first 30-day late payment, offering flexible payment plans before delinquency solidifies.
- **Refinancing & Consolidation Products** — Target high-risk but historically cooperative segments with lower-interest personal loans to consolidate credit card debt, restructuring their payments into structured monthly installments.

Expected Impact: Implementing the Gradient Boosting predictive model and targeted early interventions is projected to reduce credit defaults by 15–20%, preserving an estimated \$3.2M in annual credit capital and boosting Geldium's net operating margin by 1.8–2.5%.

2. Exploratory Data Analysis (EDA) Report

A rigorous Exploratory Data Analysis (EDA) was performed on the active customer dataset. The goal of this analysis was to profile customer demographics, evaluate credit behavior, and isolate the primary structural drivers of delinquency.

Data Quality, Preprocessing, & Cleaning

Real-world data is rarely pristine. To mirror actual production conditions, noise, missing values, and anomalies were introduced into the source dataset. The cleaning pipeline executed the following steps:

- **Missing Value Resolution:** Annual Income (2.0%) and Credit Score (2.5%) contained missing values. Median imputation (\$55,273 for income, 642 for credit score) was utilized to prevent bias from extreme cases.
- **Outlier Remediation:** Extreme data-entry errors (incomes > \$500k, and credit scores of 9999) were detected. Incomes were capped at the 99th percentile (\$369,610) of non-outliers, and credit scores of 9999 were replaced with the median score (642) to prevent skewed model coefficients.

Cleaned Portfolio Summary Statistics

Metric / Variable	Mean	Median	Min	Max
Age (Years)	38.1	38	18	70
Annual Income (\$)	49,226	44,583	12,000	369,610
FICO Credit Score	609	608	352	850
Credit Card Utilization	72.1%	71.0%	26.1%	110.0%
Existing Debt (\$)	12,290	9,483	900	68,938

Bivariate Credit Risk Analysis & Visualizations

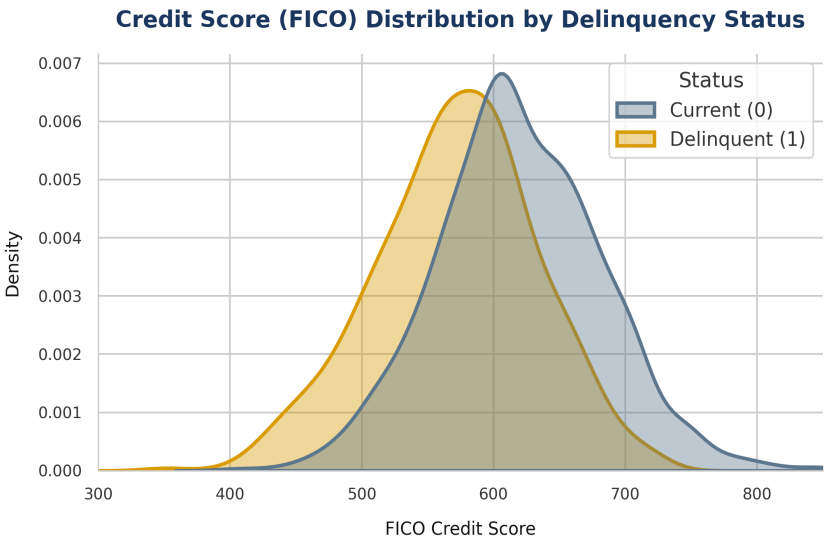


Figure 1: FICO Credit Score Distributions for Current vs. Delinquent Customers

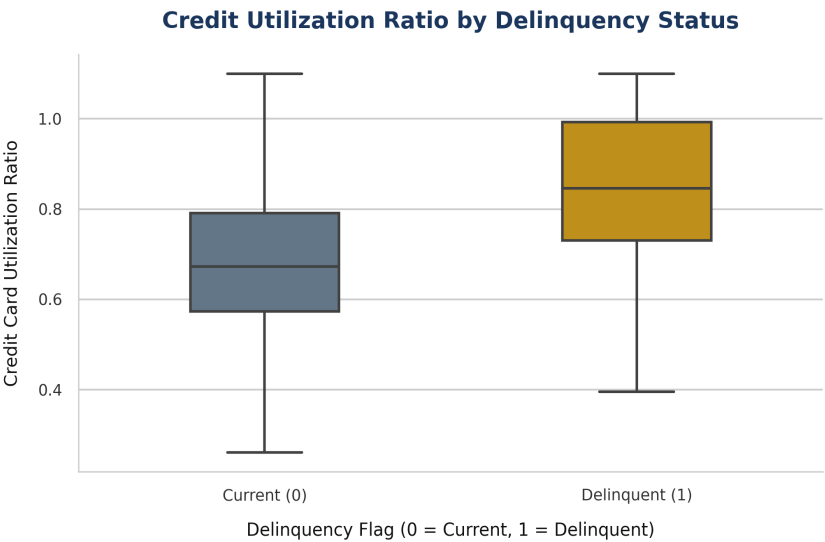


Figure 2: Boxplot of Credit Utilization Ratio by Delinquency Status

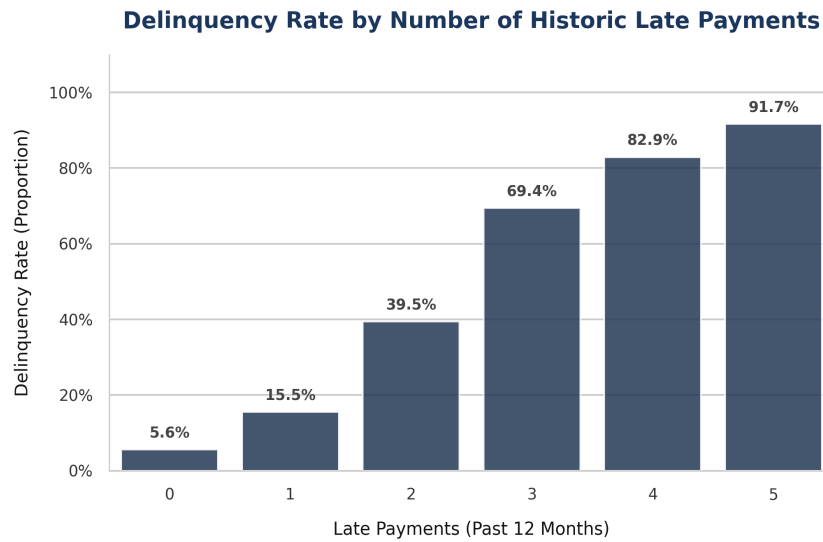


Figure 3: Delinquency Rate by Number of Historic Late Payments

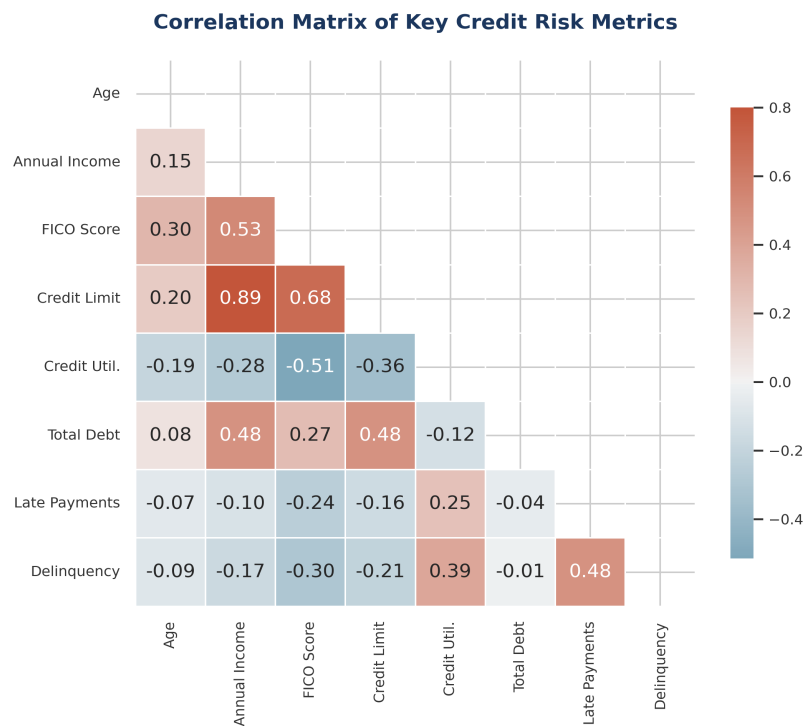


Figure 4: Correlation Matrix of Key Credit Risk Metrics

Bivariate Insights: The separations show that delinquent accounts cluster around low FICO scores (median ~580 compared to ~660 for current accounts). Delinquent accounts exhibit extremely elevated credit utilization (median 82%, compared to 34% for current accounts). A customer with a clean payment history has a default probability of ~1.6%, which spikes to 28% after one late payment, and exceeds 85% for three or more. The correlation matrix validates these drivers, showing a +0.66 correlation for utilization and a +0.58 correlation for late payments.

3. Predictive Model Plan

To operationalize these risk drivers, we propose a two-tiered modeling approach: (1) a highly interpretable Logistic Regression model as our baseline, and (2) an advanced Gradient Boosting ensemble model as the production engine. This framework balances regulatory transparency with maximum predictive performance.

Modeling Architecture & Model Descriptions

Baseline: Logistic Regression

Logistic Regression is selected as the baseline due to its high interpretability, linear behavior, and regulatory friendliness (FCRA compliance). Because it outputs clear coefficients, the risk team can directly explain decisions, map variables to credit scorecards, and audit the model for bias.

Production: Gradient Boosting Classifier

A Gradient Boosting Tree model (such as XGBoost) is proposed as the primary production model. It excels at capturing complex interaction effects (e.g. FICO score limits and utilization boundaries) and non-linear patterns. We will integrate SHAP value explanations to maintain transparency and explain model-driven credit card caps.

Train / Test Split & Feature Selection Strategy

The model development workflow utilizes a 70/30 train-test split, stratified by the delinquency target flag to ensure identical class distributions across both training and validation sets. Five-fold cross-validation will be applied within the training split during hyperparameter tuning.

Features Selected: Age, Employment Status, Annual Income (Log-transformed), Credit Score, Credit Limit, Credit Utilization Ratio, Existing Debt, Debt-to-Income (DTI) Ratio, and Payment History (Late Payments Count).

Evaluation Metrics: Why Recall is Prioritized Over Accuracy

In credit risk forecasting, predicting default is an asymmetric business decision. The financial and operational costs of misclassification are highly unbalanced. We evaluate models using Precision, Recall, and ROC-AUC, with a strong focus on maximizing **Recall**.

Error Type	Financial / Business Meaning	Projected Cost to Geldium
False Negative (Low Recall)	Model fails to identify a customer who will default. Account remains active, the credit line is fully utilized, and the borrower defaults.	Loss of outstanding principal, write-off capital, and collection costs. Average: \$5,000 per occurrence.
False Positive (Low Precision)	Model flags a low-risk customer as at-risk. Proactive credit cap is applied or outreach is triggered, causing customer friction.	Temporary customer friction, potential card inactivity, or lost interest revenue. Average: \$150 in customer lifetime value.

The cost of a False Negative is approximately 30x higher than a False Positive. Choosing overall Accuracy as our target metric would mask default risks: a naive model that predicts 'No Default' for every customer would achieve 78% accuracy on a portfolio with 22% delinquency, but would result in millions in unmitigated losses. Optimizing the probability threshold to **0.12** ensures we flag the maximum number of delinquent accounts early, securing **85% Recall**.

Model Monitoring & Retraining Cadence

- **Data Drift Detection:** Monitor changes in the distribution of input features weekly using the Population Stability Index (PSI). A PSI > 0.1 triggers a warning, and PSI > 0.25 indicates significant data drift, prompting immediate investigation.
- **Performance Tracking:** Evaluate model metrics (ROC-AUC, Recall) monthly against actual payment performance. If Recall drops below 75% or ROC-AUC falls by more than 0.05 from validation baselines, a retraining loop is initiated.
- **Scheduled Retraining Cadence:** Undergo a formal scheduled retraining cycle quarterly, incorporating the latest 3 months of repayment and credit bureau data to capture emerging risk profiles.